

Intelligent Light Project Concept

Overview

We are about to build intelligent bike light for late night bikers. Its purpose is to prevent incoming bikers from getting blinded by the bike light which might lead to a disaster. The mechanism will detect an incoming object and decrease the light brightness, as well as lower its position to provide a clear view for the user. Besides the bike light, in the bundle, additional turn signal lights are added which can be mounted on each side of the steering wheel to more visibly show your moving direction while biking. Both lights are attached to a button which when hold, flashes to the adequate side.

Components

- **Microcontroller:** Adafruit ESP32-S3 Feather (Includes a high-quality JST-PH battery connector)
- **Distance sensor:** TFmini Plus (€45) / TF-Luna (€25)
- **Side lights:** NeoPixel strip x8 WS2812 (4 LEDs each side)
- **Front light:** NeoPixel ring x7 WS2812 (Cool white color)
- **Shifter (component that boosts the voltage):** Adafruit 74AHCT125 Quad Level-Shifter
- **Nighttime sensor:** TEMA6000
- **Front light movement (servo):** MG90S micro servo
- **Rechargeable Battery:** Lithium Polymer Battery 3.7V / Power Bank
- **Buttons (turn signal lights):** Motorcycle dual handlebar switch
- **Prototyping board (to solder everything into one):** Adafruit perma-proto half-sized breadboard
- **Box:** 3D printed packaging
- **Additional components:** wires, resistors, capacitors, heat shrink tubing